Worksheet 9.1

# Identifying and classifying minerals

Student: ..... Class: .....

## Identifying minerals

**1.** Two common rocks are composed of the minerals shown in the images at right. Limestone contains calcite, and sandstone contains quartz.



Calcite



Quartz

The following table provides information about the properties of these minerals.

| Mineral | Colour | Streak (colour of powdered mineral) | Hardness | Lustre |
|---|---|---|---|---|
| Calcite | Colourless, white | White | 3 | Dull to vitreous |
| Quartz | Colourless, purple, red, yellow | White | 7 | Vitreous |

(a) Explain how calcite crystals can be distinguished from quartz crystals.

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(b) Pure quartz crystals are colourless. Explain the large variation in colour indicated in the table.

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## Creating a classification key

**2.** Use the following table to create a written classification key for these minerals.

| Mineral | Colour | Streak | Hardness | Lustre |
|---|---|---|---|---|
| Chalcopyrite | Yellow | Green, black | 3.5–4 | Metallic |
| Chlorite | Grey | White, green | 2.2–5 | Pearly |
| Galena | Grey | Grey | 2.5 | Metallic |
| Haematite | Grey | Red, brown | 5–6 | Metallic |
| Magnetite | Black | Black | 5.5–6.5 | Metallic |
| Pyrite | Yellow | Green | 6–6.5 | Metallic |

**Key:**

**1A.**

**1B.**

Calcite photo: madllen / 123RF.com
Quartz photo: thomaseder / 123RF.com

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